

The functional convergence as a substitute for dividends and debt: Evidence from Brazil

Abstract

We test if the Novo Mercado (NM) – a new segment of shares for Brazilian companies that voluntarily commit to greater requirements of corporate governance – has influenced the use of dividends and debt, which are two well-known mechanisms used to mitigate agency problems between companies and minority shareholders in Brazil. We find that the presence of firms in the NM reduces the use of dividends and debt, probably due to the belief that the additional requirements of the NM improve the corporate governance and reduce the possibility of expropriation of minority shareholder's wealth. That is, the results suggest that NM is a market-level effective trust-generating mechanism of corporate governance that can make up for country-level poor legal protection of investors.

Keywords: *Capital structure, Corporate governance; dividends; corporate debt; Brazil.*

1. Introduction

Previous literature in finance has emphasized the importance of dividends and corporate debt to alleviate the conflicts between controlling and minority shareholders (Jensen and Meckling, 1976; Jensen, 1986; Rozeff, 1982; Drobetz et al. 2010; Pindado et al. 2012). According to this perspective, distributing more dividends or increasing the leverage in companies reduces the free cash flow that otherwise could be used for purposes that are detrimental to the interests of minority shareholders. At the same time, the *Law and Finance* approach (see La Porta et al. 1997, 1998) link firms' financial policies to the legal protection of the investors. That is, if laws are effective in the protection of investors, financial markets are broader and more valuable and investors are willing to pay more for financial assets. In this sense, La Porta (1997, 1998) find that the legal origin of the countries is of great importance for the protection of minority shareholders. Further, they find that common-law

origin countries tend to protect investors. Consistently with this approach in common-law origin countries there is a higher distribution of dividends and investors are willing to pay more for debt.

However, paying more dividends and keeping higher levels of debt forces managers to depend on external financing for future projects. If the securities market is under-developed or if it is under financial constraints, the firm's cost of external financing will increase (Rozzef, 1982). This is the case of Brazil, a French-civil-law origin country with many problems related to the protection of minority shareholders. Previous literature describes Brazil as a country with historic high private benefits of control and both weak legal rules and firm-level governance (Black et al. 2012), high ownership concentration (Gorga et al. 2010), and inherent problems of information asymmetry (Moreiras et al. 2012). Nevertheless, in recent years, many reforms have been undertaken in order to improve Brazilian securities market and make it more attractive for investors. Most centrally, BM&FBovespa (the main Brazilian stock exchange) promoted a voluntary reform in which three segments with different requirements of corporate governance were launched: the *Nível* (Level) 1, the *Nível* (Level) 2 and the *Novo Mercado* (New Market) (NM). According to De Paula (2009), both *Nível* 1 and *Nível* 2 were created to give firms time to progressively improve their corporate governance aiming gradually to meet the requirements of NM –the level with the most rigorous exigencies of corporate governance. Companies that voluntarily subscribe to NM have to proceed according to its corporate governance requirements, which leads their shares to a higher valuation. This reform has led to mitigate the information asymmetry and improve the protection of minority shareholders.

This voluntary reform is based on the concept of "functional convergence" which proposes that the improvement in the protection of minority shareholders of countries with weak corporate governance must occur at the securities market level due to the difficulties to reform the laws in these countries (Coffe, 1999). However, the literature presents controversial views on the effectiveness of this concept. Johnson and Shleifer (2000) point out that in the absence of strong laws, efficient judicial systems, or a fair regulator, there is no way to protect investors because of the low possibility of enforcing contracts or laws.

In this context, we address two main research questions: (1) Do the requirements to which companies commit in the NM really mitigate the negative impact of weak investor protection at the country-level in Brazil? (2) Do Brazilian listed firms use the NM as a substitutive device to the use of dividends and debt as trust-generating mechanisms?

To address these two research questions, we first investigate whether the NM has an outcome or a substitution effect on dividend policies and if it has impacted on the use of debt of Brazilian listed firms. Then, we analyze the effect of NM on the relationship between dividends and debt. The reasoning is that dividends and debt are alternative mechanisms of corporate governance (Jensen, 1986) and if the NM really acts as an alternative monitoring mechanism, it will strengthen the negative relationship between debt and dividends. Finally, we test the effect of the NM on the distributions of dividends for firms with high investment opportunities. The reasoning is that if the NM acts as a trust-enhancing mechanism, shareholders may require less dividends in presence of investment opportunities since they trust firms will not expropriate their wealth through negative net present value projects.

Our results show a negative relationship between NM and both dividends and debt levels. We also find that NM strengthens the negative relation between dividends and debt. The results for the relationship between NM firms and the payment of dividends in presence of investment opportunities is not significant. These evidences confirm that firms use the NM as a trust-enhancing mechanism able to substitute both dividends and debt and it allows them to preserve its flexibility to finance new projects instead of disgorging their cash as dividends.

This study contributes in several ways to the corporate governance literature. First, we show that corporate governance in Brazil works as a commitment device used by companies to express the intention of not expropriating minority shareholders. Second, we contribute to the importance of the “functional convergence” in the security markets. That is, our paper shows that in the Brazilian context market-level reforms are effective in promoting meaningful changes in companies’ policies and contribute to the improvement of the protection of minority shareholders. Third, our evidence shows that the institutional framework of the countries may evolve over time, which challenges the view that considers the protection of minority shareholders a result of a static legal origin, unable to change over

the years. Finally, our evidence in an emerging market can contribute to other countries with similar institutional features as a mechanism to enhance the corporate governance and to enhance the development of their market shares, by improving the confidence of the investors. Since most of the literature is based on UK and US data relatively little is known about dividend and debt policies in emerging markets. Therefore, we add to the existent literature by providing empirical evidence for an emerging market.

The remainder of the paper is organized as follows. In section II, we describe the characteristics of Brazil's institutional framework and the Bovespa's corporate governance lines of shares and the previous literature on dividends policy, debt and investment opportunities in order to identify the relation between them and to propose the influence of the Brazilian context in such topics. In section III, we introduce the data and the specific empirical models used to test our hypothesis. In section IV we present the descriptive analysis and the results of the explanatory analysis. In the last section, we mark the main conclusions achieved.

2. Literature Review and Hypothesis

2.1. Brazilian institutional framework

Previous literature describes Brazil as having a low level of shareholders' protection, a low level of development of its financial market in which companies face financial constraints to access to external capital (Crisóstomo et al. 2014a). Aldrighi and Oliveira (2007) find that there is a very high concentration of vote and cash flow rights in the hands of controller shareholders, and a predominant presence of family controlled firms.

Notwithstanding, Brazil has been object of important structural reforms in the last decades. Legislative adjustments to strengthen the protection of external shareholders were implemented, followed by the process of privatization of public companies in the end of the last century, which aided in the modernization of key sectors of the economy. Macroeconomic reforms also led to the reduction of the inflation, reached through the *Plano*

*Real*¹, which, associated with internal changes in Brazilian main stock market (The Sao Paulo Stock Exchange), attracted new investors and contributed to the development of Brazilian stock market (Crisóstomo et al. 2014b).

However, such adjustments were not effective enough to remove obstacles whose overcoming is essential for the complete development of its stock market. Problems such as the high interest rates and impediments related to the protection of minority shareholders persisted and limited the access to external capital, which kept Brazilian companies heavily dependent on banks. In order to ensure the rights of minority shareholders and reduce the asymmetry information in the market, The Sao Paulo Stock Exchange² used its power of self-regulation³ to launch in 2000, the Novo Mercado (NM). The companies that voluntarily subscribe to the NM are required to adopt more stringent practices of corporate governance, following important principles to the effective functioning of markets. Bovespa also launched *Nível 1* (level 1) and *Nível 2* (level 2), both intermediary segments with less exigent requirements of corporate governance than NM. The goal of these segments was to give firms time to improve their corporate governance practices in a progressive way until they reach the conditions to subscribe on the NM (De Paula, 2009).

The main differences between these lines of shares are described in Table 1. In sum, according to the Brazilian legislation (Law 10,303 / 2001), the listed companies must keep at least 50% of their shares with voting rights. The NM requires firms to issue only ordinary shares. By law, there is no definition of a minimum free float for listed companies. The NM requires companies to keep at least 25% of free float⁴. The NM also imposes greater representation

¹ The Plano Real (Real Plan) was a set of measures taken to stabilize the domestic currency in nominal terms. It was enacted in 1994 and included the introduction of a new currency followed by a series of contracting fiscal and monetary policies.

² In 2008, the Sao Paulo Stock Exchange (Bovespa Holding S.A.) and the Brazilian Mercantile and Futures Exchange (BM&F S.A.) merged. The new entity was called “*Nova Bolsa*”. Then, its name was changed to “BM&FBovespa”.

³ The Law 6,385 / 1976, §4⁰, Art.21, provides the power of self-regulation for the Sao Paulo Stock Exchange. Therefore, the requirements of its corporate governance levels are not just recommendations (soft law). Instead, the noncompliance with these requirements can result in the punishment of the defendants applied by the arbitral chamber.

⁴ Free float represents all the shares issued by the Company, except those ones held by the controlling shareholder or by people related to him, by the company's management, treasury shares and special class preferred shares.

and independence in the board of directors. Brazilian legislation requires companies to keep at least 3 directors and does not require them to be independent. The NM obliges the presence of at least 5 permanent members on the boards of which at least 20% must be independent directors. The national legislation requires a tag along⁵ of 80% in case of control alienation. In the NM, companies must guarantee a tag along of 100% in transferences of control. The NM also requires new transparency measures with the adoption of IFRS and US GAAP. The NM also enplaned the arbitration chamber, an entity endowed with power to arbitrate on demands of minority shareholders.

Moreover, *Nível 1*, *Nível 2* and *Novo Mercado* are all obligated to perform at least one annual meeting between analysts and shareholders, to provide the company's annual calendar, to disclosure contracts signed between the company and related parties and to disclosure transactions made by controlling shareholders as well.

Donaggio (2012) points that the voluntary reform of BM&Fbovespa aimed to attract foreign capital to boost the development of the Brazilian stock market. For this purpose, it was necessary to reduce the perception of risk for investors through the implantation of good corporate governance rules. The requirement of issuance of only ordinary shares in NM aimed to provide a fair treatment for minority shareholders, following the principle of "one share - one vote". The requirement of a tag along of 100% was intended to reduce the disparity of rights between majority and minority shareholders, since, at that moment, the Brazilian legislation allowed the issuance of 2/3 of preferred shares. A minimum free float of 25% aims to improve the liquidity of the market. The arbitral chamber was created in order to resolve conflicts by experts in corporate matters in a faster way.

Table 1 - BM&FBovespa Corporate Governance Levels

Item	Traditional Market	Nível 1 (Level 1)	Nível 2 (Level 2)	Novo Mercado
Proportion of voting rights	Voting and non-voting shares are allowed	Voting and non-voting shares are allowed	Voting and non-voting shares are allowed	Only voting right shares
Minimum free float	There is no rule	Minimum of 25% of outstanding shares	Minimum of 25% of outstanding shares	Minimum of 25% of outstanding shares

⁵ Tag Along is a device that assures that if the majority shareholder sells his stake, minority shareholders have the right to sell their shares at the same conditions applied to controlling shareholders.

Board of Directos	At least, three members	At least, three members	At least, three members	At least, five effective members and at least 20% obligatory independent directors.
Disclosure standards	ITRs, DFPs and IANs.	More requirements for ITRs, DFPs and IANs.	IFRS or US GAAP	IFRS or US GAAP
Tag Along	80%	80%	100% for ordinary shares and 80% for preferred shares	100%
Arbitration chamber	Optional	Optional	Required	Required

Notes: the table presents the main differences between the levels of corporate governance launched by BM&Fbovespa. The Traditional Market represents the exigencies for companies that choose not to list in any corporate governance level. ITRs (quarterly reports), DFPs (standardized financial statements) and IANs (annual reports) are required for public companies by Brazilian legislation. In N1, these reports must inform the contracts signed between companies and related parties and also transactions made by controlling shareholders. IFRS is the International Financial Reporting Standard and US GAAP is Generally Accepted Accounting Principles adopted by U.S. Securities and Exchange Commission (SEC). Tag along are rights that allow minority shareholders to sell his shares in similar conditions to the controller shareholders deals. The arbitration chamber is a mechanism created in order to solve conflicts between minority shareholders and companies or controller shareholders in a quickly and confidentially way through specialized referees. Companies of N2 and NM are obligated to adopt it while companies of TM and N1 have the option to adopt it or solve their conflicts in the common justice.

2.2. Dividends and the Novo Mercado

Many studies have researched the effect of corporate governance on dividends decisions. For example, La Porta et al. (2000) raise two alternative hypotheses for the relationship between dividend policy and shareholder rights. The first one states that the presence of an effective legal system allows shareholders to obtain higher levels of dividends; in other words, dividends are an outcome of effective legal protection of shareholders (the outcome hypothesis). The second one poses that an ineffective legal system can lead companies to distribute dividends ratios more meaningful as a way to replace the weak investor protection (the substitution hypothesis). These authors find that firms operating in countries with better legal protection of shareholders distribute higher dividend payout compared to their counterparts located in worse minority legal protection countries, confirming the outcome hypothesis.

Many other papers support the outcome hypotheses through the adoption of country-level or firm-level corporate governance practices in payout policies. For instance, Djankov et al. (2008) construct an index to measure the legal protection of minority shareholders against expropriation in 72 countries and confirm that in countries whose laws come from the

common-law model, due to the quality of transparency in the negotiations and emphasis on litigation *ex-ante*, there is greater protection against expropriation compared to countries based on civil-law legal system, which also influences the way companies conduct their policies in both systems.

Farinha (2003) find a link between dividend policy and firms compliance with Cadbury Code of Best Practices. The author argues that better governed firms use dividend policies more intensely and that the legal environment and dividend policy complement each other. In a related study, Adjaooud and Ben-Amar (2010) underline a positive relation between the governance score and the payout policy in a sample of Canadian firms. Francis et al. (2011) also find a positive relationship between an index of internal governance, the adoption of antitakeover laws and dividend distribution.

On the other hand, many articles confirm the substitution hypothesis in environments other than the UK and the USA. In this regard, Chae et al. (2009) find that companies with higher financial constraints reduce dividends even in presence of agency costs as their corporate governance becomes better. John et al. (2015) also find that firms with weaker corporate governance pay more dividends, which is attributed to the role of dividends in mitigating manager-shareholders agency conflicts aggravated by weak corporate governance. Again, Fatemi and Bildik (2012) find that firms in Civil-law countries distribute more dividends than those in countries with higher shareholder protection. In a study with Arab markets, Athari et al. (2016) find that Islamic banks use dividends policies as a substitute mechanism for reducing agency problems.

O'Connor (2006) supports that the protection of minority shareholders in U.S regime reduces the demands for dividends. Specifically, he finds that firms listed in trading cross-listing Level 2/3 ADR substitute dividends for enhanced bondings. In this sense, Lam et al. (2012) with a sample of Chinese firms, proves that both foreign ownership and cross-listing (firms listed in more transparent and rigorous markets) have significant negative effects on cash dividends. The authors attribute this fact to the effectiveness of these mechanisms in reducing tunneling activities. Brockman and Unlu (2009) find that the substitution model is related to bad disclosure environments. That is, in countries with higher information assymetry, firms

pay higher dividends. Finally, Mitton (2004) finds that, for Civil-law countries, dividends and corporate governance have a strong negative relation.

To this same extent, in the Brazilian context, we expect that the different levels of corporate governance between the NM and the traditional market (MT) affect the level of payout distributions of Bovespa listed firms. In this way, two possible situations may arise: in one scenario, companies distribute more dividends as an outcome of their compliance with NM corporate governance requirements. Alternatively, it is always possible to occur a reduction trend in dividends payments might occur as an effect of the generation of trust between companies and minority shareholders. Our first hypothesis, thus, is as follows:

H1a: NM companies disgorge more cash in the form of payout policy than those of the traditional market.

H1b: NM companies disgorge less cash in the form of payout policy than those of the traditional market.

2.3. Debt policy and the Novo Mercado

In their pioneering study of the agency theory, Jensen and Meckling (1976) argued that debt works as a disciplinary mechanism by forcing managers to have fixed obligations. Additionally, Jensen (1986) argued that debt forces companies to disgorge cash flow that, otherwise, could be used to finance negative net present value projects. Agrawal and Knoebet (1996) emphasize the importance of the lenders for monitoring companies' activities since their capital is affected by borrowers' performance.

However, part of the previous literature on debt links the effectiveness of debt to the quality of the governance practices. Clearly, if capital market institutions are ineffective in inhibiting the opportunistic behavior, the expropriation via debt can be facilitated by allowing blockholders to control more resources without diluting their control stake. That is, an ineffective corporate governance can incentive an environment favorable to expropriation through related-party transactions and pyramidal structures (Faccio, 2001; Setia-Atmaja et al. 2009, Ruiz-Mallorquí & Santana-Martín, 2011). Moreover, increasing the quality of financial reporting can lead credit providers to reduce debt interests in the loans since they

have more information to assess the borrower's conditions to honor its commitments, which can reduce the cost of the debt (Chen & Zhu 2013).

On the contrary, if capital market institutions are effective, two possibilities may arise. First, higher levels of debt can reduce the agency conflicts between controlling and minority shareholders. That is, debt can act as a disciplinary mechanism for managers. From this perspective, in presence of good corporate governance, companies would prefer to keep higher levels of debt for signaling their intention of not expropriating investor's wealth.

Alternatively, previous literature finds that better corporate governance facilitates firms' access to external capital, which positively impacts on the development of the financial markets and also on the growth of the firms as a consequence of a reduction in the financial constraints (Claessens & Yurtoglu, 2013). In turn, less financially constrained firms use less debt since they have greater access to external capital (Kaplan and Zingales, 1997; Whited and Wu, 2006; Francis et al. 2013, Florackis et al. 2015). Consequently, if the corporate governance reduces information asymmetry, both shareholders and managers would prefer firms to reduce debt levels, since the corporate governance replaces the disciplinary function of debt and keeps the freedom that less leveraged companies have to handle a higher cash holding.

Hence, we pose that, in the Brazilian context, listing in the NM may raise two alternative hypotheses. On one scenario, NM firms can increase the debt levels of firms due to its effectiveness to inhibit the expropriation of controlling shareholders by debt. Alternatively, it is possible that the trust-enhancing provided by the NM reduces the debt levels by facilitating firms access to external capital and as well as the preference of insiders to discharge from the obligations of the debt.

H2a: NM firms are more leveraged than non-NM firms

H2b: NM firms are less leveraged than non-NM firms

2.4. The Influence of the *Novo Mercado* in the Relation Between Dividends and Debt

Likewise, according to the agency theory, debt and dividends have a negative association since both can be considered as alternative monitoring mechanisms. Basically, they have the

same function of disciplining managers and avoiding the expropriation of minority shareholders' wealth. Farinha (2003) attributes it to the debt covenants and related restrictions imposed by debtholders that force managers to reduce dividends in presence of debt. Agrawal and Knoeber (1996) points out that the use of debt is important for reducing agency problems because it induces monitoring by lenders. Therefore, if the company is leveraged, it may naturally reduce the use of dividends. Confirming this assertion, Mulyani et al. (2016) in an Indonesian sample (an emerging civil law country) find a strong negative relation between debt and the distribution of dividends and attributes this to the use of debt instead of dividends to reduce agency problems in that country.

In the Brazilian case, we assume that the NM is effective in ensuring transparent accounts and protecting the rights of minority shareholders, which makes the debt decisions to be taken due to the concerns of informed external suppliers of capital and not as a consequence of the interests of the controlling shareholders. Therefore, we pose that the NM acts as an alternative monitoring mechanism that increases the negative relationship between debt and dividends.

H3: NM companies distribute less dividends in presence of debt.

2.5. The Influence of the NM in the Relation Between Dividends and Investment Opportunities.

As previously explained, dividends are believed to be effective for reducing agency costs. In this stream, Easterbrook (1984) points out that dividends are effective in monitoring managers at lower costs compared to other monitoring mechanisms and may adjust the level of the risk taken by managers because they influence firms' financing policies, dissipating cash and inducing firms to float new securities. This vision is supported by previous researches in Brazil that show that the payment of dividends exerts a positive impact on the value of the firms with fewer growth opportunities (López-Iturriaga & Crisóstomo, 2010). That is, when investment opportunities are low the investors prefer to receive dividends for avoiding free cash flow problems.

But dividends also increase the transaction costs by leading firms to raise new capital to finance new projects (Easterbrook, 1984; Rozeff, 1982). According to this perspective, it is possible that higher requirements of corporate governance influence dividends policy of firms

in presence of high investment opportunities. In this sense, Michaely & Roberts (2012) find a negative association between information and agency problems and dividends in presence of investment opportunities. That is, if the corporate governance is strong, dividends payments decrease when the cash is needed. If it is true that the financial policies of companies are influenced by the concerns to reduce agency costs, in presence of an effective corporate governance, companies with low investment opportunities may naturally distribute more dividends compared to their counterparts with higher investment opportunities due to the higher risk of expropriation or overinvestment to which those companies are subject.

Previous works confirm that investors are sensitive to the importance of corporate governance in the relationship between dividends and investment opportunities. Officer (2011), for instance, finds that firms with weak corporate governance present a higher return to the initial announcement of dividends than their counterparts with strong corporate governance do. Lin & Shen (2012) in a sample of Taiwan firms, confirm that in presence of investment opportunities, in a weak corporate governance regime, firms pay substantial higher dividends as a reputation-enhancing mechanism. In contrast, if the corporate governance regime is sound, the levels of dividends are not influenced by the corporate governance.

Hence, we believe that corporate governance reduces information asymmetries that influence shareholders to reject possible profitable projects. In the Brazilian case, we pose that companies listed in the NM with high investment opportunities keep lower levels of dividends to preserve its flexibility to finance new projects.

H4: NM companies distribute less dividends in presence of investment opportunities.

3. Empirical design

3.1. Sample

Our sample comprises the listed Brazilian companies that are included on the Worldscope Thomson database for the years 2001 to 2011. This period covers the first 11 years of the NM and the years before the actual macroeconomic crisis of Brazil, which is a good sign that macroeconomic issues will not overly affect our results as in crisis period. We dropped firms

with missing values on debt, total assets and cash dividends. In addition, we omit financial firms due to their differences in the corporate structure compared to the rest of the sample.

After these conditions, our data is presented as an unbalanced panel, consisting of 205 companies and 1,831 firm-year observations. We choose to keep an unbalanced panel instead of a balance approach to mitigate survivorship bias problems. In the sample period (11 years) many cases of delists, mergers and acquisitions occurred. Selecting only firms with the same number of observations would decrease the sample to an unacceptable size. We retained firms with at least 5 consecutive years of valid data to allow for dynamic analysis.

3.2. Variables and models

In line with the previous literature dividends (DIV) is the payout ratio measured as total cash dividends paid over total assets (Kusnadi & Wei, 2011; Pindado et al. 2012; Lozano & Caltabiano, 2015). The cash holding (CASH) is measured as the sum of cash and current assets over total assets (Lozano & Caltabiano, 2015). SIZE is the natural logarithm of the total assets (Kusnadi & Wei, 2011; Lozano & Caltabiano, 2015). DEBT is measured as total debt over market value of equity plus total debt plus preferred stock. The investment opportunities (IO) is the market to book ratio, measured by the sum of market value of the company, total assets and the book value of equity over total assets (Adam & Goyal, 2008). We use ROA as a measure of profitability (Pindado et al. 2008). ROA is measured by earnings before interests and taxes over total assets. Investments (INV) is measured by capital expenditures over total assets. TANG represents the tangible assets. It is measured as tangible fixed assets over total assets (Alonso et al. 2005).

Regarding the models used, we include in them a dummy variable NM, whose coefficient allows us to test the effect of *Novo Mercado* on the dependent variables. Some firm characteristics remain constant over the time and are unobservable for the researcher. For instance, dividends and debt are strongly linked to specific characteristics of firms, such as firm culture, firm life cycle and the propensity of managers to pay dividends and to take risks. These effects were included in the models through the variable η_i , which is then eliminated by first differences.

The regressions can be affected by time-varying macroeconomic effects, as the macroeconomic set is not constant over the time. We controlled this effect using a time dummy variable, d_t . I_i is an industry dummy variable that represents the effect of the industry effect on our dependent variable because dividends and debt are strongly related to company's specific business area. Finally, other factors not included in the mentioned variables are captured by the random disturbance, ϑ_{it} . Then, our first model is:

$$DIV_{it} = \beta_0 + \beta_1 DIV_{it-1} + \beta_2 DEBT_{it} + \beta_3 ROA_{it} + \beta_4 IO_{it} + \beta_5 CASH_{it} + \beta_6 SIZE_{it} + \beta_7 INV + \beta_8 NM + d_t + \eta_i + I_i + \vartheta_{it}. \quad \text{Eq. (1)}$$

To test the hypothesis 2, we replace the dependent variable of Eq.(1) for the variable DEBT. We add to the model other variables recognized in previous literature by their influence in debt policies, as showed in Eq. (2):

$$DEBT_{it} = \beta_0 + \beta_4 IO_{it} + \beta_4 TANG_{it} + \beta_5 SIZE_{it} + \beta_6 INV + \beta_7 ROA_{it} + \beta_8 NM + d_t + \eta_i + I_i + \vartheta_{it}. \quad \text{Eq. (2)}$$

To test the hypothesis 3, we interact NM with the variable DEBT. Given that NM is a dummy variable equal to 1 if the firm is listed in the *Novo Mercado* and zero otherwise, β_2 is the coefficient of debt for firms not listed in the NM, since $NM = 0$, and $\beta_2 + \alpha_1$ represents the coefficient for firms listed in the NM, since $NM = 1$. In the latter case, a test of linear restrictions is done to determine whether the sum $\beta_2 + \alpha_1$ is significantly different from zero.

$$DIV_{it} = \beta_0 + \beta_1 DIV_{it-1} + (\beta_2 + \alpha_1 NM) * DEBT_{it} + \beta_3 ROA_{it} + \beta_4 IO_{it} + \beta_5 CASH_{it} + \beta_6 SIZE_{it} + \beta_7 INV + \beta_8 NM + d_t + \eta_i + I_i + \vartheta_{it}. \quad \text{Eq. (3)}$$

Finally, to test the hypothesis 4, we changed the dummy variable in Eq. (2) to the variable IO. Therefore, β_4 is the coefficient of investment opportunities for firms not listed in the NM, since $NM = 0$, and $\beta_4 + \alpha_1$ represents the coefficient for firms listed in the NM, since $NM = 1$.

$$DIV_{it} = \beta_0 + \beta_1 DIV_{it-1} + \beta_2 DEBT_{it} + \beta_3 ROA_{it} + (\beta_4 + \alpha_1 NM) * IO_{it} + \beta_5 CASH_{it} + \beta_6 SIZE_{it} + \beta_7 INV + \beta_8 NM + d_t + \eta_i + I_i + \vartheta_{it}. \quad \text{Eq. (4)}$$

3.3. Methodology

To avoid the endogeneity problem, we estimate all models using an instrumental variable method. According to Ogaki (1993), the best method is generalized method of moments (GMM) because it embeds all other instrumental variables methods as special cases. Specifically, we used the system GMM, which was proposed by Blundell & Bond (1998) as the most reliable method. Furthermore, previous literature supports the use of system GMM to address the dynamics of dividends (Henry, 2011; Pindado et al. 2012). The system GMM uses the lagged values of the independent variables as instruments, which are highly correlated to the regressors that they instrument. Therefore, this method is very suitable to solve the endogeneity problem.

Following Blundell & Bond (1998), we use as instruments each right-hand side variable in the models, lagged one to four times for equations in differences (except for lagged dividends, whose instruments are lags two to five times) and just once in the level equations and only one instrument, t , for the equations in levels. We also controlled for the dynamic effects by including in dividends models a lag of the dependent variable $Div_{i,t-1}$. Moreover, we use the Windmeijer (2005) adjustment for finite samples. It aims to avoid potential downward bias in the estimated asymptotic standard errors and to improve the robustness of the results.

We also performed some tests to check potential misspecifications of the models. First, we used the J Hansen of over identifying restrictions to test the absence of correlation between the instruments and the error term. The results support the validity of the instruments in all models used. Second, following Arellano & Bond (1991), we use the m_2 statistic to test the lack of second-order serial correlation in the first difference residual. The results find no such problems in the models used. Finally, we use the Wald test of jointly significance of the reported coefficients, a Wald test of the joint significance of the time dummies and a Wald test of the significance of industry dummies.

4. Results

4.1. Descriptive Analysis

Table 2 – Summary statistics for the variables used in the models

Variable	obs	Mean	Standard deviation	1° quartil	2°quartil	3°quartil
DIV	1831	.028	.051	.000	.012	.033
DEBT	1831	.445	.304	.181	.422	.708
IO	1831	2.144	14.929	.701	.954	1.455
SIZE	1831	6.125	2.078	4.739	6.185	7.509
CASH	1274	.122	.129	.026	.085	.175
ROA	1820	.082	.029	.041	.104	.164
INV	1831	.054	.059	.014	.040	.076
TANG	1735	.524	.240	.339	.517	.706

Notes: this table provides the number of observations, means, standard deviations, as well as the values in first, second and third quartiles for all the variables used in the models.

Table 3 - *t*-test

Variable	All (1)	Non-Novo Mercado (2)	Novo Mercado (3)	<i>t</i> -statistic (2)-(3)
No. Obs.	1831	1564	267	
DIV	.028	.032	.027	-1.36*
DEBT	.445	.476	.261	11.005***
M/B	2.144	2.197	1.833	.0368
SIZE	6.125	5.981	6.965	-7.246***
CASH	.455	.440	.516	-4.791
ROA	.082	.079	.097	-.964*
INV	.054	.052	.071	-4.873***
TANG	.524	.545	.400	9.069***

Notes: this table provides the *t*-test for the variables used in the models. Diff. is the difference between the means of each variable between NM and MT. Column 1 shows the values for the whole sample while Columns 2 and 3 show the results separately for non-NM firms and for NM firms.

Table 3 presents a preliminary analysis of the differences between Non-NM firms and NM firms. For doing so, we conduct a test of difference of means for the variables used in the models. The results show that NM-firms pay higher dividends than non-NM firms [See column (2)-(3) *t*-statistic]. Meanwhile, since NM firms differ from their non-NM counterparts in many aspects for which a univariate analysis does not control, we cannot conclude that the differences in dividend ratios are due to differences in other firm-level characteristics.

The results on the other variables highlight some interesting characteristics of the sample. For example, the coefficient for debt is lower for NM-firms, these companies also present a higher cash-holding, which can engender substantial free cash flow, leading to conflicts of interest between shareholders and managers. Likewise, NM-firms are bigger than non-NM

firms. In larger firms it is likely that managers do not own a large controlling stake in the firm, thus increasing agency conflicts, since their interests are likely to be worst aligned with minority shareholders (Setia-Atmaja et al. 2009). Moreover, our measure of tangibility (TANG) shows that NM-firms have a lower collateral. The lower tangibility in NM assets structure decreases the guarantees in situations of bankruptcy or default situations, which contributes to these firms to be riskier. Concerning the investment opportunities variable (M/B), there are not significant differences between both types of firms.

To sum up, as exposed above, it is worth it to emphasize that NM firms gather more features to generate agency conflicts since these companies are bigger, keep higher cash holding and, in spite of presenting lower investment opportunities, they invest more than their non-NM counterparts. Therefore, it seems that the corporations use the NM as a trust-enhancing mechanism which allows them to have discretion to finance their investment opportunities.

4.2. Regression Results

The empirical evidence provided in Column 1 of Table 4 confirms that, *ceteris paribus*, companies listed in NM disgorge less cash in the form of dividends than those firms listed in the other Bovespa's levels. This negative correlation confirms our hypothesis 1b and represents a strong signal that, in Brazil, the protection of minority shareholders provided by NM reduces their demands for dividends.

The evidences for our debt model (Table 4 Column 2) show that there is a strong negative relationship between the NM and the corporate debt. This confirms our hypothesis 2b, in which, we pose that the NM facilitates firms' access to external capital and that a better corporate governance leads insiders to discharge from the obligations of the debt. Moreover, the results shown in Table 4 Column 3 suggest that the presence in the NM reduces the payment of dividends for leveraged companies. In other words, firms listed in the NM disgorge less dividends if they use debt. The coefficient for firms listed in the NM is lower [$\beta_5 + \alpha_1 = -.019 + (-.023) = -.042$, $t = -3.7^{***}$] than for non-NM firms ($\beta_5 = -.019$). This result supports hypothesis 3 and confirms the effectiveness of the *Novo Mercado* as an alternative monitoring mechanism that enforces the negative relationship between dividends and debt. Concerning the payment of dividends in presence of investment opportunities, Table 4

(column 4) Shows that in spite of the NM variable shows a negative effect on dividends, this coefficient is not significant at any level. coefficient in the relation between dividends and investment opportunities. the results for the variable is not significant at any level.

These results should be seen as corroborating the substitution model of dividends (Mitton, 2004; O'Connor, 2006; Lam et al. 2012, Lin & Shen, 2012) rather than the outcome model (La Porta et al. 2000; Farinha, 2003; Brockman and Unlu, 2009; and Adjoud and Ben-Amar, 2010; Francis et al. 2011; John et al. 2015, among many others). That is, in the Brazilian set, the adoption of rigorous requirements of corporate governance through the compliance with the NM exerts a negative effect on the distribution of dividends.

Table 4 – Estimations results

Variables	DIV (1)	DEBT (2)	DIV (3)	DIV (4)
Intercept	.006 (.011)	.174*** (.058)	.001 (.017)	-.005 (.013)
DIV_{it-1}	.513*** (.11)		.512*** (.010)	.534*** (.109)
DEBT_{it-1}		.716*** (.030)		
SIZE_{it}	.001 (.001)	.005 (.007)		.001 (.002)
IO_{it}	.007*** (.002)	-.0001 (.001)		.012** (.006)
DIV_{it}		-.497** (.220)		
DEBT_{it}	-.026*** (.008)		-.023*** (.007)	-.019*** (.006)
CASH_{it}	-.019 (.015)		-.020 (.021)	-.009 (.019)
INV_{it}	-.023 (.019)	.020 (.146)	-.030 (.025)	-.036 (.023)
TANG_{it}		-.055 (.045)		
ROA_{it}	.090* (.054)	-.047 (.063)	.091 (.056)	.085* (.047)
NM_{it}	-.010** (.004)	-.062*** (.018)		
DEBT*NM_{it}			-.019** (.008)	
IO*NM_{it}				-.005 (.004)
T			-3.7***	3.00***
Z₁	30.85***	122.97***	30.68***	34.12***
Z₂	1.27	13.36***	1.66	1.39

Z₃	1.15	0.77	1.43	1.40
M₂	1.58	-0.32	1.59	1.56
Hansen	132.07	182.75	132.53	123.33

Notes: this table shows the regressions system GMM for the robustness tests, Notes: 1) SEs are given in parentheses. 2)*, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively. 3) Z_1 is a Wald test for the joint significance of the coefficients on the explanatory variables, asymptotically distributed as X^2 under the null of no relationship, degrees of freedom in parentheses, Z_2 is a Wald test for the joint significance of the time dummies, asymptotically distributed as X^2 under the null of no relationship, degrees of freedom in parentheses and Z_3 is a Wald test for the joint significance of the sector dummies, asymptotically distributed as X^2 under the null of no relationship, degrees of freedom in parentheses. 5) M_2 is a serial correlation test of second order using residuals in first differences, asymptotically distributed as $N(0,1)$ under the null of no serial correlation. 6) The Hansen test, a test of the overidentifying restrictions, asymptotically distributed as X^2 under the null of no correlation between the instruments and the error term; the degrees of freedom in parentheses.

Overall, our results indicate that the NM not only exerts an important effect on the financial policies analyzed, but also that it has been employed as a device to substitute other corporate governance mechanisms like dividends and debt, which are used to reduce the agency problems in the companies. Our evidence shows that for NM companies, the use of debt reduces the payment of dividends. We attribute this to the effectiveness of the NM to ensure transparent accounts and to improve the rights of minority investors, which reduces the agency conflicts between insiders and minority shareholders. Therefore, the NM does work as an alternative monitoring mechanism to dividends and debt.

Likewise, our findings indicate that, the relationship between dividends and investment opportunities is not affected by the presence in the NM. It confirms previous findings in Brazil that dividends are used for signalling future investment opportunities (Iturriaga & Crióstomo, 2010; Lozano & Catalbiano, 2015). Regarding the control variables, we find a significant negative relation between dividends and debt (Columns 1 and 3 in Table 4), which confirms that both governance mechanisms substitute each other. As expected, past dividends and profitability present a positive effect on dividends (Brav, *et al.*, 2005). We also find a positive significative relation between dividends and investment opportunities. This is in accordance to previous findings in Brazil (López-Iturriaga & Crióstomo, 2010; Lozano & Catalbiano, 2015) and confirms the signalling function of dividends respect to future growth opportunities. Finally, the variables investment, size and cash do not show a significant effect on dividends. Respect our second model, Table 4 (column 2) shows that lagged debt present a positive significant effect on present debt, consistently with the adjustment of the capital structure (Ozkan, 2001; Hovakimian and Li, 2011).

4.3. Robustness Checks

We conduct some tests to verify if the voluntary reforms in Brazilian capital market were really effective and to which extent they impact on the behavior of Brazilian listed companies. First, we conduct tests with different measures of dividends, debt and investment opportunities. As an alternative measure for dividends, we use DIVCE, which is the ratio of total cash dividends paid over common equity. As for our first measure of dividends, the results confirm a negative relationship between DIVCE and NM (see Table 5 Column 1). Similarly, our alternative measure of debt (MLEV), measured by the total debt over the sum of total assets plus the book value of equity plus market value of equity, confirmed the previous findings that the NM has a negative relationship with debt (see Table 5 Column 2). Likewise, Table 5 Column 3 confirms that NM strengthens the negative relationship between dividends and debt [since $\beta_5 + \alpha_1 = -.024^{**} + (-.041^{**}) = -.065$, $t = -3.31^{***}$, bigger than the coefficient for MLEV ($\beta_5 = -.024$)]. To measure investment opportunities, we use Tobin's Q, represented by the ratio of market value of equity plus book value of debt over the sum of book value of equity plus book value of debt (Alonso et al. 2005), as an alternative measure for M/B. Confirming our previous findings, the variable $IO * NM_{it}$ is not significant. Therefore, the previous findings were confirmed in all the alternative variables used even for dividends, debt and between dividends and debt and dividends and investment opportunities.

Second, we test our models with a two-step instrumental variable method. In short, the results for the first hypothesis confirm the negative impact of the presence in the NM on dividends and debt (Table 6 column 1 and 2). Furthermore, the results show the negative effect of the NM on the relationship between dividends and debt [$\beta_5 + \alpha_1 = -.021^{*} + (-.012^{*}) = -.033$, lower than for MLEV ($\gamma_1 = -.021^{*}$)]. As before, the coefficient of interaction between investment opportunities and the dummy NM do not show a significant coefficient (Table 6 column 4).

Table 5 – Estimations results - Robustness checks with different measures for the dependent variables.

Notes: this table shows the regressions system GMM for the robustness tests, Notes: 1) SEs are given in parentheses. 2)*, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively. 3) Z_1 is a Wald test for the joint significance of the coefficients on the explanatory variables, asymptotically distributed as X^2 under the null of no relationship, degrees

Variables	DIVCE (1)	MLEV (2)	DIVCE (3)	DIVCE (4)
β_0 Constant	-.088*** (.033)	.0677* (.040)	-.007 (.010)	.024* (.013)
β_1 DIV _{it-1}	.350*** (.111)		.510*** (.108)	.569*** (.100)
β_1 MLEV _{it-1}		.731*** (.033)	-.024** (.009)	
β_2 SIZE _{it}	.003 (.005)	.008 (.005)	.002 (.002)	-.0005 (.001)
β_3 IO _{it}	.034*** (.010)	.0002 (.0003)	.007*** (.002)	.0003 (.0006)
β_4 DIV _{it}		-.274** (.123)		
β_5 DEBT _{it}	.0002 (.030)			-.030*** (.006)
β_6 CASH _{it}	.039 (.046)		-.020 (.017)	-.011 (.014)
β_7 INV _{it}	.051 (.087)	.060 (.095)	-.016 (.027)	-.032 (.021)
β_8 TANG _{it}		-.045 (.033)		
β_9 ROA _{it}	.186* (.100)	-.003 (.027)	.093* (.047)	.087 (.059)
β_{10} NM _{it}	-.028** (.012)	-.027** (.013)		
α_1 MLEV. NM _{it}			-.041** (.018)	
α_2 IO. NM _{it}				-.0005 (.000)
t			-3.31***	-0.16
Z_1	10.44***	110.24***	25.75***	35.53
Z_2	1.83*	12.00***	1.59	1.12
Z_3	2.44**	0.71	1.10	1.48
M_2	1.60	0.45	1.60	1.61
Hansen	128.34	188.11	133.34	140.02

of freedom in parentheses, Z_2 is a Wald test for the joint significance of the time dummies, asymptotically distributed as X^2 under the null of no relationship, degrees of freedom in parentheses and Z_3 is a Wald test for the joint significance of the sector dummies, asymptotically distributed as X^2 under the null of no relationship, degrees of freedom in parentheses. 5) M_2 is a serial correlation test of second order using residuals in first differences, asymptotically distributed as $N(0,1)$ under the null of no serial correlation. 6) The Hansen test, a test of the overidentifying restrictions, asymptotically distributed as X^2 under the null of no correlation between the instruments and the error term; the degrees of freedom in parentheses.

The purpose of our additional checks is to ensure the accuracy of our previous findings and check if the behavior of NM firms is a causality effect. That is, if the companies in the NM have some features that make them distribute less dividends and use less debt in their capital structure. Our findings confirm that the corporate governance has been effective in

substituting debt and dividends in their well-known functions of reducing the agency problems in the firms.

Table 6 – Estimations results – Robustness checks with instrumental variables method.

Variables	DIV (1)	DEBT (2)	DIV (3)	DIV (4)
Intercept	.017* (.009)	.074*** (.027)	-.004 (.011)	-.003 (.011)
β_1 DIV_{it-1}	.541*** (.054)		.588*** (.057)	.586*** (.061)
β_1 DEBT_{it-1}		.861*** (.022)		
β_2 SIZE_{it}	.000 (.001)	.003 (.004)	.000 (.001)	.000 (.001)
β_3 IO_{it}	.005*** (.002)	-.005 (.003)	.004** (.001)	.005** (.002)
β_4 DIV_{it}		.460 (.461)		
β_5 DEBT_{it}	-.017*** (.001)		-.012* (.006)	-.013* (.006)
β_6 CASH_{it}	-.010 (.008)		-.010 (.008)	-.009 (.009)
β_7 INV_{it}	-.063** (.032)	.385** (.159)	-.046 (.033)	-.049 (.033)
β_8 TANG_{it}		-.048 (.041)		
β_9 ROA_{it}	.164*** (.046)	-.498* (.290)	.142*** (.050)	.147*** (.053)
β_{10} NM_{it}	-.007** (.003)	-.025* (.015)		
α_1 DEBT*NM_{it}			-.021* (.012)	
α_2 IO*NM_{it}				-.003 (.002)
Obs.	1097	1313	1097	1097
Firms	156	200	156	156
F	61.58	162.57	63.16	62.42
R²	0.6084	0.6874	0.6055	0.6014
Adj. R²	0.6317	0.6836	0.5967	0.5925
Year	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes

Notes: this table shows the regressions system GMM for the robustness tests, Notes: 1) SEs are given in parentheses. 2)*, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively. Obs. And Firms represent the number of observations and the number of firms for each regression. F represents the joint significance of the estimators. The R² and the adj. R² represent the coefficient of determination of the models. Year and Industry denote the presence or not of dummies of years and industries in the models.

5. Conclusions

The aim of this research was to analyze the efficacy of a market-level reform led by the Brazilian capital markets to reduce the historical problems of the Brazilian law concerning the protection of minority shareholders. To this end, we examine the impact of NM on dividends and debt policies. The rationale is that dividends and debt are both mechanisms used by companies to mitigate agency conflicts. We find that NM firms distribute less dividends and keep lower levels of debt than non-NM firms. Consequently, we argue that companies use the NM as a trust-generating mechanism because shareholders believe that it reduces the information asymmetry and that it mitigates concerns about minority shareholders wealth expropriation for Brazilian listed firms. Our results are in line with a substitution model of dividends, according to which a better corporate governance substitutes for dividends payments (Mitton, 2004; O'Connor, 2006; Lam et al. 2012, Lin & Shen, 2012). Likewise, our results reveal that NM companies are less leveraged than non-NM companies. Several reasons can explain this finding. Firstly, NM firms can reduce the role of debt as a disciplinary mechanism since the NM works as a substitutional disciplining device. Secondly, it is possible that NM reduces the possibility of expropriation through related party loans because of its requirements of transparency. Moreover, like for dividends, it is possible that the reputational effect of the NM alleviates the financial constraints faced by its listed firms. If it is true, NM firms naturally resort less to debt to finance their activities.

Additionally, we use a dynamic dividend model to verify the effect that the NM exerts on the relationship between dividends and debt. The results show that more leveraged companies listed in the NM pay even less dividends than non-NM companies, which confirms that shareholders trust that NM inhibits the expropriation through related-party loans of pyramidal structures of ownership. Going a step further, we analyse the effect of NM on dividends in presence of investment opportunities. The evidence confirms that, in presence of investment opportunities, NM firms payout less dividends than non-NM firms. We argue the reason for this is that the NM reduces information asymmetries that influence shareholders to reject possible profitable projects. That is, minority shareholders understand that the NM reduces the possibility of managers to dress up the projects.

This research provides new insights in the corporate governance area by showing that the “functional convergence” can work as an effective complementary mechanism to the

country-level deficiencies in protecting minority shareholders. That is, changes in regulations at the level of securities markets can improve the corporate governance of the countries. Moreover, the results provide evidence that the institutional environment of the countries may evolve over time, which reduces the role of the legal origin as the main determinant of the quality of the protection of minority shareholders in the countries.

An additional insight of this paper regards the institutional setting. Specially, we provide evidence that applies to a setting other than UK and US, since literature on corporate governance and finance in emergent markets is scarce. So, we add to this line of research by analyzing how a voluntary reform in an emerging market affects specific firms decisions, such as dividends distributions and debt levels.

Similarly, our findings can be useful for practioners and policy makers. Our results indicate that investors can benefit of the enlistment of a firm in the NM because it will lead to a reduction in the information assymetries between firms controllers and minority shareholders. For policy makers, our results indicate that efforts to improve the corporate governance on the securities markets can boost the participations of investors and reduce the difficulties to access the extern capital.

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